
REWARDING THE GRAPH BEHIND THE CHAIN: TOPOLOGY-AWARE PROCESS SUPERVISION FOR RL AND REASONING DISTILLATION

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ABSTRACT

Long chain-of-thought reasoning is not purely linear: intermediate results are reused across non-adjacent steps, while unsupported conclusions may refer backward or form circular support. Outcome rewards and step-local process rewards do not expose these failures. We propose TopoPRM, a topology-aware implicit process reward model that first recovers an order-agnostic directed support graph from a free-form trace and then projects it onto a dependency DAG. Crucially, the backward and cyclic edges removed by this projection remain separate direction-consistency and acyclicity signals. We aggregate outcome, format, direction, acyclicity, and continuity with a topology-conditioned Choquet reward whose interactions explicitly encode structural agreement rather than averaging the two graph diagnostics into a generic topology score. The same projected graph guides distillation toward a compact dependency-preserving trace, without requiring graph-format generation from the policy. We evaluate whether TopoPRM advances the fixed-scale accuracy–length Pareto frontier over outcome-only GRPO and additive or multiplicative aggregation, together with controlled edge-recovery, topology-perturbation, and reward-hacking tests.

1 INTRODUCTION

Reinforcement learning with verifiable rewards (RLVR) has substantially advanced long chain-of-thought (CoT) reasoning in large language models and now serves as a core mechanism for training mathematical and symbolic reasoning ability (Wei et al., 2022; Kojima et al., 2022; DeepSeek-AI, 2025; Yeo et al., 2025; Shao et al., 2024; Ke et al., 2026). From sparse final-answer rewards (Cobbe et al., 2021; DeepSeek-AI, 2025) to widely used process reward models (PRMs) that attach supervision to intermediate positions or prefixes (Lightman et al., 2023; Wang et al., 2024; Uesato et al., 2022), the dominant training interface still represents a trace as a *linear sequence of steps*. In practice, long reasoning is rarely linear: in mathematical derivations, a later conclusion often depends on several non-adjacent premises, an intermediate equation is reused by multiple downstream derivations, and subgoals branch before merging into the final answer (Besta et al., 2024; Zhong et al., 2026). Reconciling this structural mismatch has emerged as a core challenge for advancing reasoning supervision beyond linear credit assignment.

Recent reasoning models have made the macroscopic structure of inference clearer and more reliable, yet they remain confined to the CoT-as-result view. DeepSeek-R1 (DeepSeek-AI, 2025), DAPO (Yu et al., 2025), and Tulu 3 (Lambert et al., 2024) demonstrated that outcome-only rewards scale, but they could not distinguish a well-supported derivation from a trace that reached the same answer through unsupported intermediate claims, and they remained vulnerable to reward hacking (Gao et al., 2023; Skalse et al., 2022). On the PRM side, denser signals have been obtained through Monte Carlo rollouts (Wang et al., 2024; Luo et al., 2024), trained verifiers (Zhang et al., 2025; Khalifa et al., 2025; Zhao et al., 2026), and implicit outcome-derived scoring (Yuan et al., 2025; She et al., 2025). These methods can condition on a full prefix, but they do not explicitly recover the **multi-parent support relation** in which one intermediate result supports, or is supported by, several non-adjacent steps (Wang et al., 2025b). Likewise, token-level distillation (Agarwal et al., 2024) does not require a compact target to preserve those cross-step dependencies.

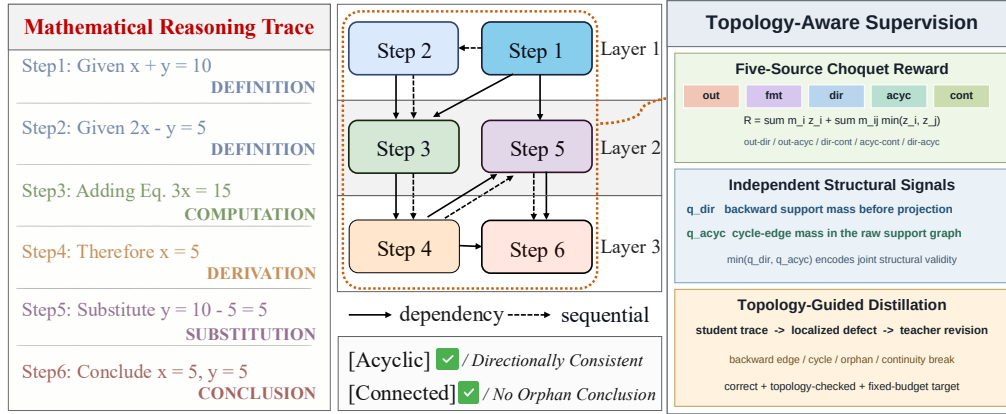


Figure 1: TopoPRM overview. A free-form mathematical trace is mapped to a layered dependency graph; direction and acyclicity enter the topology-conditioned multi-source reward, while the same graph diagnostics localize dependency-preserving teacher revisions for distillation. The policy itself never emits graph tags.

We propose **TopoPRM**, which supervises reasoning through its *implicit topological dependency structure* rather than assuming that surface order is already valid (Figure 1). A frozen, order-agnostic edge encoder first recovers a raw directed support graph $\mathcal{H}_y$ from an unchanged free-form trace. Projecting $\mathcal{H}_y$ onto a text-order DAG $\mathcal{G}_y$ makes the discarded structure informative: backward-edge mass measures direction inconsistency, while cycle-edge mass measures circular support. We retain these as separate reward sources and aggregate them non-additively with outcome, format, and continuity. In particular, direction–acyclicity synergy rewards traces that satisfy both local support order and global non-circularity, while their outcome and continuity interactions make topology change rollout ordering rather than act as a small generic bonus. Finally, the same projected graph guides distillation toward a compact dependency-preserving trace. The policy never has to decode graph tags or a special DAG format. Our contributions are threefold:

1. **Raw support graph before acyclic projection.** We recover directed dependencies without imposing textual order, and turn the backward and cyclic structure removed by projection into explicit process supervision.
2. **Topology-conditioned multi-source optimization.** We introduce a monotone 2-additive Choquet reward that keeps direction and acyclicity independent and explicitly models their mutual, outcome, and continuity interactions, then reuse the graph diagnostics for Topology-Guided Distillation.
3. **Accuracy–efficiency evaluation.** We compare against outcome-only and explicit length-shaped GRPO, projected-graph controls, additive and multiplicative aggregation, and distillation baselines under a fixed-scale, fixed-budget protocol, reporting both accuracy and the accuracy–length Pareto frontier.

2 RELATED WORK

Reinforcement Learning with Verifiable Rewards. RLVR replaces preference models with programmatic correctness checks on tasks whose answers are objectively verifiable (Shao et al., 2024; DeepSeek-AI, 2025; Lambert et al., 2024; Zeng et al., 2023). Recent work refines GRPO-style optimization through dynamic sampling and clipping (Yu et al., 2025), extended training schedules (Liu et al., 2025), finer credit assignment (Kazemnejad et al., 2025), asymmetric handling of negative samples (Zhu et al., 2025), entropy regularization (Cui et al., 2025), and simplified baselines (Ahmadian et al., 2024; Hu, 2025). Outcome-only signals nevertheless remain structurally blind: traces with very different dependency support can be reward-equivalent (Chen et al., 2026), and reward hacking persists even when the visible chain appears benign (Gao et al., 2023; Skalse et al., 2022; Wang et al., 2026b). *TopoPRM* injects frozen topology and continuity diagnostics from latent rea-

soning graphs, targeting structural credit assignment directly rather than through a correctness surrogate.

Concise-reasoning objectives introduce a separate confound: penalizing length inside a group-relative reward can assign negative advantage to a correct but long rollout. DRPO addresses this by decoupling the length-driven target distribution from correctness discrimination (Li et al., 2026a). DeCS localizes the necessary reasoning prefix with a separate detector, applies token-level reward only to the resulting chunks, and schedules prompt difficulty (Jiang et al., 2026); we reproduce its released 7B checkpoint under our fixed evaluator. LASER-D instead assigns step length rewards using targets that adapt to training-time behavior and per-question difficulty (Liu et al., 2026b); ARLCP coordinates difficulty-calibrated length and detected reflection penalties (Yu et al., 2026); and Self-Aligned Reward favors query-specific responses through a relative-perplexity signal rather than explicit trace structure (Han et al., 2026). Step-Tagging monitors semantic step types with lightweight classifiers and stops generation from their running counts (Belkhiter et al., 2025); it controls inference behavior but does not recover directed support among steps. These methods target redundancy or response efficiency, whereas our direction and acyclicity scores distinguish support relations that can occur at equal length and with unchanged wording. For multi-source reward integration, fair and stable reward composition alternates between mirror-descent weight updates and policy updates but retains a dynamic weighted sum (Li et al., 2024); HERO stratifies dense reward-model scores by binary verifier outcomes and reweights prompts by within-group variance (Tao et al., 2026). The former adapts source weights without representing source complementarity, while the latter preserves correctness ordering and makes dense feedback explicitly correctness-first. Our hypothesis is different: raw direction and acyclicity should remain identifiable, interacting reward sources rather than only additive terms or tie-breakers inside correctness strata. We therefore include additive and verifier-stratified controls under the same completion budget, and treat accuracy and length as co-primary outcomes.

Process Reward Models for Reasoning. PRMs score intermediate steps to address the sparsity of outcome rewards, originally relying on dense human annotation (Uesato et al., 2022; Lightman et al., 2023). Subsequent work reduces labeling cost via Monte Carlo or tree-search rollouts (Wang et al., 2024; Luo et al., 2024; Zhang et al., 2024; Lu et al., 2024; Zhang et al., 2025; Guan et al., 2025), generative PRMs make verifier reasoning explicit (Khalifa et al., 2025; Zhao et al., 2026; She et al., 2025), while recent process rewards use intrinsic semantic cues or explicitly tagged meta-reasoning actions (He et al., 2026; Zhang et al., 2026c). Process advantage verifiers tie step values to rollout-induced returns (Setlur et al., 2025; Lai et al., 2024); implicit PRMs derive them from outcomes alone (Yuan et al., 2025; Snell et al., 2024), and CRM conditions each step on the full preceding sequence while linking it to the final outcome (Zhang et al., 2026b). Recent benchmarks expose persistent weaknesses on fine-grained error categories and inter-step consistency (Song et al., 2025; Zheng et al., 2025). Such scorers can model a prefix but do not recover an explicit many-to-many support relation among steps. *TopoPRM requires no tagged policy format and trains no scalar neural process verifier; it derives frozen global-structure supervision from a recovered dependency graph together with local continuity.*

Structured and Graph-Based Reasoning. PARC links a completed chain to preceding premises for error identification (Mukherjee et al., 2025), while GRiD generates knowledge/reasoning nodes and verifies each node against declared parents (Wen et al., 2025). Graph of Verification adaptively decomposes a trace into a DAG and verifies node blocks in topological order without training (Fang et al., 2026). Other methods require graph-formatted reasoning: DAG-Math evaluates graph-level logical closeness (Zhang et al., 2026a), Structured Reasoning rewards within-graph flow and cross-sample overlap (Dong et al., 2026a), Structural CoT trains executable computation graphs (Anonymous, 2026b), VeriGraph makes a data-analytic agent emit a heterogeneous evidence DAG and rewards its integrity and coherence (Jin et al., 2026), the Graph Reasoning Paradigm emits labeled symbolic graphs (Liu et al., 2026a), and GoT-R1 reinforces parsability, sparsity, and non-linearity of serialized reasoning graphs with an additive reward (Li et al., 2026b). AtomGraph instead learns a graph-attention verifier over reasoning chains that have already been parsed into DAGs (Karmore, 2026). Contract-checked graph editing enforces acyclicity as a hard gate for composing candidate DAGs during inference-time search (Li & Cao, 2026). TUR-DPO sanitizes extracted claim-support graphs and linearly combines valid-path, cycle, dangling-node, and contradiction terms to shape offline preference optimization (Abdullah et al., 2026); it neither retains edge-direction residuals

nor separates direction from cyclic support. RewardFlow merges observed agent trajectories into an environment-state transition graph and propagates terminal reward along action edges (Feng et al., 2026); it does not recover support direction within one free-text derivation. FlowTracer instead derives token credit from an attention-induced answer-flow DAG (Dong et al., 2026b). Two methods named Graph-GRPO propagate credit over a predefined e-commerce relevance graph or optimize sampled multi-agent communication topologies, rather than dependencies within one free-form trace (Che et al., 2026; Cang et al., 2026). TopoPRM instead leaves policy traces in free text and recovers an unconstrained semantic-support graph before projection. Declared-parent, attention-DAG, learned-DAG, valid-DAG, and hard-gated representations rule out or do not score backward and cyclic semantic support; we retain these residuals as direction and circular-justification signals.

Post-hoc graph probes support non-sequential reasoning (Zhong et al., 2026; Matsutani et al., 2026); CoT2Graph uses recovered graphs for failure analysis, outcome prediction, and Best-of- N selection (Anonymous, 2026a), and Reasoning Consensus merges several recovered DAGs for inference-time structural ensembling (Parulekar et al., 2026). SARL clusters reasoning steps into latent functions and rewards clustering and short paths in their *undirected* transition graph, explicitly abstracting away directional control flow (Wang et al., 2026c). This small-world prior is complementary to our semantic-support relation: our direction score detects backward dependency mass, while acyclicity detects global self-support. Hidden-state cycles can reflect useful recurrent visits (Minegishi et al., 2025), while LoopBench studies self-reinforcing textual repetition during generation (Duan et al., 2026). Our cycles instead denote circular semantic support in the committed derivation: we do not penalize exploration, revision, or repeated wording, only a final support relation in which a claim depends on itself. The novelty is therefore not “reasoning as a graph,” but raw direction/cycle diagnostics as online, auditable reward sources without graph-formatted generation.

Distillation and Reasoning Compression. Knowledge distillation transfers teacher capability (Hinton et al., 2015); student-generated variants reduce train–test mismatch (Gu et al., 2024; Agarwal et al., 2024; Ko et al., 2024), including ExOPD, BOND, KDRL, and black-box GAD (Yang et al., 2026; Sessa et al., 2024; Xu et al., 2025; Ye et al., 2025). Long-CoT compression uses pruning, length penalties, or budget control (Xia et al., 2025; Luo et al., 2025; Arora & Zanette, 2026; Aggarwal & Welleck, 2025; Muennighoff et al., 2025); graph-based pruning removes weak branches from projected DAGs (Yuan et al., 2026), while Reasoning Scaffolding distills semantic operations (Wen et al., 2026). TopoPRM instead uses raw direction/cycle violations to elicit and filter teacher revisions, preserving dependencies that token budgets or post-projection pruning cannot identify. *Topology-Guided Distillation is evaluated by the accuracy–length frontier and independently audited topology preservation.*

3 METHOD

TopoPRM (Figure 2) is a topology-aware post-training framework whose stages share a graph encoder $\mathcal{E}_\phi$ and an acyclic projection Π . Stage I produces parsable reasoning traces. Stage II applies GRPO to a topology-conditioned multi-source reward. Stage III performs Topology-Guided Distillation, in which the Stage-II teacher revises student rollouts using defects localized on the projected graph.

3.1 PROBLEM SETUP

Let $x = (q, c)$ pair a mathematical problem q with optional context c , and let π_θ generate a trace $y = (s_1, \dots, s_T)$. The frozen encoder produces a raw directed support graph $\mathcal{H}_y = \mathcal{E}_\phi(y) = (\mathcal{V}, \mathcal{A}_H)$, where $i \rightarrow j$ means that step s_i supplies a result, claim, sub-goal, or assumption used by s_j . Unlike a DAG constructed only from pairs $i < j$, $\mathcal{H}_y$ may contain backward edges and cycles. We retain these violations and separately form $\mathcal{G}_y = \Pi(\mathcal{H}_y)$ by keeping support edges consistent with the trace order. Thus $\mathcal{G}_y$ supports DAG operations, while the projection cost of $\mathcal{H}_y$ exposes structural failures that would otherwise be removed by construction.

Scope of the acyclicity assumption. Acyclicity is normative only for the *semantic support relation in the committed trace*. It does not forbid exploratory computation, checking an earlier result, or revising a failed branch. Such behavior may revisit a latent state or repeat a concept while still

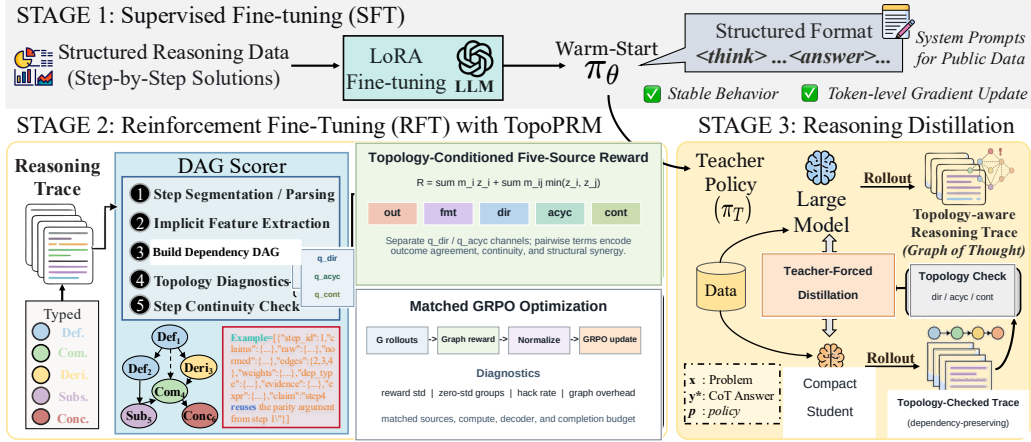


Figure 2: Overview of TopoPRM and Topology-Guided Distillation. An order-agnostic encoder recovers a raw directed support graph; its acyclic projection provides the training DAG while discarded backward and cyclic edges remain reward signals. A topology-conditioned Choquet reward aggregates multiple sources, and the projected graph guides dependency-preserving distillation.

yielding an acyclic final justification. A cycle in $\mathcal{H}_y$ has the narrower meaning that a written claim ultimately supports itself; q_{acyc} targets this circular justification rather than recurrent reasoning in general.

3.2 TOPOLOGY-AWARE DATA PROCESSING

Raw-Graph Extraction and Projection. Given a rollout y , $\mathcal{E}_\phi$ proceeds in three stages. **(i) Segmentation and parsing.** It splits y into steps and sub-questions, then extracts normalized expressions, claim keys, variables, and step types. **(ii) Order-agnostic support encoding.** Auditable rules propose high-precision symbolic candidates, while a lightweight pair encoder predicts one of $\{i \rightarrow j, j \rightarrow i, \emptyset\}$ for each unordered intra-component pair. The encoder is trained offline from semantic-teacher edges and frozen during policy optimization; a separately annotated set calibrates edge precision and direction. Retained encoder edges carry their calibrated confidence, while rule edges use unit weight. Semantic predictions override an order-based rule when both judge the same pair, preventing artificial two-cycles caused by conflicting extractors. **(iii) Acyclic projection.** We preserve all predicted edges in $\mathcal{H}_y$, set $\mathcal{G}_y = \Pi(\mathcal{H}_y)$ by removing $i \rightarrow j$ with $i \geq j$, and add an adaptive weak sequential edge only when a node remains unsupported. Full rules and abstention thresholds are in Appendix C.

Topology and Continuity Diagnostics. Let $\mathcal{A}_H^{\text{dep}}$ denote the dependency edges of $\mathcal{H}_y$, let $w_{ij} \geq 0$ be the confidence weight of edge (i, j) , and set $W_H = \sum_{(i,j) \in \mathcal{A}_H^{\text{dep}}} w_{ij}$. Let $\mathcal{A}_{\text{cyc}}$ contain the dependency edges whose endpoints lie in the same cyclic strongly connected component (size greater than one, or a self-loop), and let c_H be the fraction of nodes incident to at least one dependency edge. We first compute d_H , the forward-edge mass, and u_H , the cycle-edge mass, then shrink both diagnostics toward 0.5 when edge coverage is low:

$$q_{\text{dir}}(\mathcal{H}_y) = c_H d_H + (1 - c_H)/2, \quad d_H = \frac{\sum_{(i,j) \in \mathcal{A}_H^{\text{dep}}} w_{ij} \mathbb{1}[i < j]}{W_H}, \quad (1)$$

$$q_{\text{acyc}}(\mathcal{H}_y) = c_H(1 - u_H) + (1 - c_H)/2, \quad u_H = \frac{\sum_{(i,j) \in \mathcal{A}_{\text{cyc}}} w_{ij}}{W_H}. \quad (2)$$

When $W_H = 0$, we use $d_H = 1$ and $u_H = 0$; because $c_H = 0$ for a multi-step empty graph, both coverage-adjusted scores are nevertheless 0.5, not perfect. (A singleton trace has no possible cross-step violation and uses $c_H = 1$.) The continuity score q_{cont} measures whether each step is traceable to the question or a supported prefix of $\mathcal{G}_y$. Direction and acyclicity remain independent properties of the raw graph rather than being averaged into one topology score or checked tautologically after

projection. They are related but not redundant: every directed cycle contains a backward edge under the textual order, whereas a backward edge need not close a cycle, and raw graphs with identical backward-edge mass can have different cycle-edge mass. Thus q_{dir} detects local order inconsistency while q_{acyc} detects global circular support.

Topology-Conditioned Multi-Source Aggregation. Let $\mathbf{z} = (r_{\text{out}}, r_{\text{fmt}}, q_{\text{dir}}, q_{\text{acyc}}, q_{\text{cont}}) \in [0, 1]^5$. A weighted sum assumes that each source has the same marginal value regardless of the others. Instead, we use a monotone 2-additive Choquet model with Möbius coefficients (Choquet, 1954; Grabisch, 1997)

$$r_{\text{total}}(x, y) = \sum_i m_i z_i + \sum_{(i,j) \in \mathcal{I}_{\text{topo}}} m_{ij} \min(z_i, z_j), \quad \mathcal{I}_{\text{topo}} = \{(o, d), (o, a), (d, c), (a, c), (d, a)\}. \quad (3)$$

Here o, d, a, c denote outcome, direction, acyclicity, and continuity. All $m_i, m_{ij} \geq 0$ and sum to one, so the reward is bounded and monotone. The five interactions assign additional value when each raw-graph diagnostic agrees with outcome or continuity and when direction agrees with acyclicity. This makes topology a constituent of cross-source utility rather than a correctness-first bonus; no length source enters the main reward. We ablate the capacity against equal-weight, Shapley-matched additive, and multiplicative alternatives, and remove the direction–acyclicity interaction in isolation.

3.3 TOPOLOGY-GUIDED POST-TRAINING

Stage I: Supervised Warm-Start. We perform Supervised Fine-Tuning (SFT) on structured mathematical reasoning data to teach the policy to produce stepwise solutions parsable by $\mathcal{E}_\phi$. This reduces invalid rollouts during subsequent reinforcement learning, since topology diagnostics are meaningful only when the trace contains recoverable reasoning steps.

Stage II: GRPO with TopoPRM Rewards. Starting from the warm-start checkpoint, we optimise the policy with GRPO. For each completion, $\mathcal{E}_\phi$ constructs $\mathcal{H}_y$ and $\mathcal{G}_y$; Eq. 3 then scores the completion. Standard group normalization acts on the complete Choquet reward, so topology can change the ordering of rollouts rather than only break ties inside an outcome stratum. Reward-hacking tests separately measure whether structurally plausible but answer-incorrect traces are over-ranked.

Stage III: Topology-Guided Distillation. The optimized model after Stage II serves as the teacher π_T , while a smaller pretrained checkpoint initializes the student π_S . Topology-Guided Distillation conditions target construction on topology diagnostics from student-generated traces. For each prompt x , the student first rolls out $y_{\text{init}} \sim \pi_S(\cdot | x)$; $\mathcal{E}_\phi$ produces $\mathcal{H}_{y_{\text{init}}}$ and $\mathcal{G}_{y_{\text{init}}}$; a dispatcher ψ then issues a revision instruction targeting a backward edge, cyclic component, orphan conclusion, or continuity break. The teacher generates a revised target under this instruction. Thus topology determines where supervision is requested and which revisions are retained, rather than serving only as a post-hoc score.

Distillation objective. Let $A(y_{\text{init}}, y_{\text{rev}})$ indicate that the revised trace has a correct boxed answer and valid format, satisfies $q_{\text{dir}}^{\text{rev}} \geq \tau_d$, $q_{\text{acyc}}^{\text{rev}} \geq \tau_a$, and $q_{\text{cont}}^{\text{rev}} \geq \tau_c$, does not reduce either raw-graph diagnostic relative to the initial trace, and fits a fixed token budget B . The retained dataset is $\mathcal{D}_{\text{distill}} = \{(x, y_{\text{rev}}) : A(y_{\text{init}}, y_{\text{rev}}) = 1\}$. The student minimises standard teacher-forced distillation loss

$$\mathcal{L}_{\text{distill}}(\theta) = -\mathbb{E}_{(x,y) \sim \mathcal{D}_{\text{distill}}} \sum_{t=1}^{|y|} \log \pi_S(y_t | x, y_{<t}). \quad (4)$$

Unlike static trace distillation, the target distribution is concentrated on revisions elicited by the current student’s structural failures. The explicit direction and acyclicity constraints prevent apparent compression from being obtained by deleting necessary support or introducing circular justification. The full pipeline is summarised in Appendix Algorithm 1.

4 EXPERIMENTS

Draft audit status. Red cells are not claims. Each method uses one canonical seed-0 training run; reported intervals quantify evaluation-set uncertainty, not training variance. The original EMNLP tables remain archival in Appendix G until they satisfy the canonical protocol below.

4.1 CANONICAL PROTOCOL

Fixed model scale and budget. The primary comparison starts from the same Qwen3.5-9B Stage-I checkpoint and changes only the Stage-II reward. The public frontier is restricted to dense released checkpoints with 7B–9B parameters; sparse models with similar total but much smaller active parameter counts are outside this matched scale. All variants use rank-64 LoRA, 300 GRPO steps, four generations per prompt, learning rate 5×10^{-6} , $\beta=0.04$, and a 4,096-token completion budget. Stage I uses 18,487 structured math solutions for training and 983 source-disjoint solutions for validation, three epochs, learning rate 5×10^{-5} , and global batch 32. Stage II uses 14,893 unique public train-split questions after exact normalized-question exclusion against every primary evaluation set. We remove 76 of 14,969 leakage-free candidates whose exact Qwen3.5 chat prompts exceed 512 tokens, rather than truncating their problem statements. Reference-solution DAGs are withheld because their step indices are not aligned with generated traces; q_{dir} , q_{acyc} , and q_{cont} are computed only from each rollout’s $\mathcal{H}_y$ and $\mathcal{G}_y$. The pair encoder is frozen in every topology-aware run; the 9B semantic teacher is used only to create its offline supervision and never appears in the GRPO reward loop.

Benchmarks and metrics. The primary fixed-scale frontier uses GSM8K, MATH-500, the official 675-item English text-only mathematics subset of OlympiadBench, and AIME 2024, spanning elementary through competition mathematics. Pass@1 exact-answer accuracy and mean visible response tokens are co-primary; correctness is read only from the last non-empty box or a terminal explicitly marked answer, so intermediate expressions in truncated traces receive no credit. EOS padding is excluded by retokenizing each decoded response without special tokens. Method *A* Pareto-dominates *B* only when it is no less accurate and no longer, with one strict inequality; pass@5 and maj@5 are secondary diagnostics. Every reproduced row uses one seed-0 sampled response per item with temperature 0.6, top- $p=0.95$, top- $k=20$, min- $p=0$, repetition penalty 1.0, identical task content and verifier, and a 4,096-token cap. Released chat templates and documented role/thinking controls are retained; matched rows share tokenizer and message roles (Appendix A.4).

Statistics and provenance. Every row maps to a checkpoint, manifest, and per-example output. Marginal benchmark accuracy uses Wilson 95% intervals; token means and stratified macros use item bootstrap, while matched accuracy and length differences use paired bootstrap over shared items. With one training run, these quantify evaluation-set uncertainty only. Public checkpoints with longer budgets or repeated decoding are re-evaluated under our single-response 4,096-token protocol rather than quoted in Table 1. The human audit reports agreement, edge precision/recall, direction accuracy, and abstention coverage separately from teacher-held-out validation.

4.2 ACCURACY–LENGTH FRONTIER (RQ1)

Table 1 tests the central empirical claim: at a fixed 9B adaptation scale and response budget, does the raw-directed TopoPRM reward move the accuracy–length frontier beyond Stage-I SFT, outcome-only GRPO, explicit outcome–format–length shaping, the dynamic difficulty-aware LASER-D baseline, and matched additive or multiplicative multi-source rewards? Our matched LASER-D adaptation preserves the released mechanism: training-time length coverage determines a target for each rollout-difficulty band, and only correct responses within that target receive its step reward (Liu et al., 2026b); Appendix B records the budget-scaled thresholds and monitor protocol. The completed Stage-I row has macro accuracy 42.1% (stratified item-bootstrap 95% interval [40.2, 44.4]) and 526 visible tokens ([423, 647]); these intervals quantify evaluation-set uncertainty, not training variance. A SOTA claim will be made only if TopoPRM is non-dominated against all matched rows and improves the reproduced accuracy–length frontier over the listed 7B–9B public checkpoints. Until then, the table remains a preregistered comparison rather than a result narrative. *External bars.* FlowTracer uses an attention-induced answer-flow DAG under a 1K-output protocol (Dong et al.,

Table 1: Fixed-budget accuracy–length comparison at dense 7B–9B scale. Cells report pass@1 (%) and mean visible response tokens from one seed-0 sample per item under matched sampling, task content, and a 4,096-token cap; released chat templates and required thinking controls are retained. Adapted methods use one canonical seed-0 training run. **Accuracy intervals are Wilson; token and paired-difference intervals use item bootstrap. Neither measures training variance.**

Method	GSM8K		MATH-500		OlympiadBench		AIME 2024		Macro avg.	
	Acc.	Tok.	Acc.	Tok.	Acc.	Tok.	Acc.	Tok.	Acc.	Tok.
<i>Public checkpoints, reproduced with the same evaluator</i>										
Qwen3-8B (Qwen Team, 2025)	91.4	1,948	64.6	3,185	–	–	0.0	4,096	–	–
SpyRL-Qwen3-8B-Math (Wang et al., 2026a)	–	–	–	–	–	–	–	–	–	–
GoT-R1-8B (Li et al., 2026b)	95.2	357	–	–	–	–	16.7	2,018	–	–
DeepSeek-R1-0528-Qwen3-8B (DeepSeek-AI, 2025)	–	–	–	–	–	–	–	–	–	–
DeCS-7B (Jiang et al., 2026)	–	–	–	–	–	–	–	–	–	–
MiMo-7B-RL-0530 (Xiaomi LLM Core Team, 2025)	–	–	–	–	–	–	–	–	–	–
Nemotron-Cascade-8B-Thinking (Wang et al., 2025a)	–	–	–	–	–	–	–	–	–	–
Klear-Reasoner-8B (Su et al., 2025)	–	–	–	–	–	–	–	–	–	–
OpenReasoning-Nemotron-7B (NVIDIA, 2025)	83.0	2,429	51.2	3,260	–	–	0.0	4,096	–	–
AceMath-RL-Nemotron-7B (NVIDIA, 2026)	–	–	–	–	–	–	–	–	–	–
Qwen3.5-9B (Qwen Team, 2026)	76.4	2,710	40.4	3,627	–	–	0.0	4,096	–	–
<i>Matched Qwen3.5-9B adaptation from the same Stage-I checkpoint</i>										
Stage-I SFT	85.8	137	59.6	283	19.6	604	3.3	1,080	42.1	526
Outcome-only GRPO	–	–	–	–	–	–	–	–	–	–
Outcome + format + length GRPO	–	–	–	–	–	–	–	–	–	–
LASER-D (matched) (Liu et al., 2026b)	–	–	–	–	–	–	–	–	–	–
Matched additive reward	–	–	–	–	–	–	–	–	–	–
Matched multiplicative reward	–	–	–	–	–	–	–	–	–	–
TopoPRM (Choquet)	–	–	–	–	–	–	–	–	–	–

Table 2: Direction and acyclicity mechanism tests. **Top:** matched policy ablations use the canonical decoder and budget. **Bottom:** frozen-encoder falsification on the same 120 traces (56 cycle-closure pairs); deltas are perturbed minus source and uncertainty uses paired trace bootstrap. The final row tests cancellation from premature source averaging. Bottom-panel values are measured diagnostics, not policy results. **Red cells await canonical policy outputs.**

	GSM8K		MATH-500		OlympiadBench		AIME 2024		Macro		Generated topology	
Policy variant	Acc.	Tok.	Acc.	Tok.	Acc.	Tok.	Acc.	Tok.	Acc.	Tok.	q_{dir}	q_{acyc}
Full TopoPRM	–	–	–	–	–	–	–	–	–	–	–	–
w/o direction source	–	–	–	–	–	–	–	–	–	–	–	–
w/o acyclicity source	–	–	–	–	–	–	–	–	–	–	–	–
w/o raw-graph sources (projected DAG only)	–	–	–	–	–	–	–	–	–	–	–	–
Rules only (no pair encoder)	–	–	–	–	–	–	–	–	–	–	–	–
w/o continuity	–	–	–	–	–	–	–	–	–	–	–	–
Frozen-encoder falsification tests on identical source traces												
Trace perturbation	n	p_{real}	Δq_{dir}	Δq_{acyc}	$\Delta \text{Cov.}$	$\Delta \text{Back.}$	ΔCycle	Targeted failure mode				
Reverse all steps	120	–	–.758	+ .059	+ .068	+ .834	– .028	Textual order / direction				
Matched cycle closure	56	–	.000	– .339	.000	.000	> 0	Circular support at fixed backward mass				
Random edge deletion ($p = .10$)	120	.072	– .013	– .004	– .030	–	– .011	Missing dependencies				
Random direction flip ($p = .30$)	120	.295	– .216	– .078	.000	–	+ .084	Orientation corruption				
Random false-edge addition ($p = .30$)	120	.261	– .084	– .220	+ .029	–	+ .242	Spurious and cyclic support				
Premature source-aggregation diagnostic on reversed controls												
Trace subset	n	Corr.	σ_{dir}	σ_{acyc}	σ_{avg}	$\sigma_{\text{avg}}/\sigma_{\text{dir}}$	Opposite Δ	Interpretation				
Reversed traces	120	– .483	.142	.088	.063	.443	30.0%	Acyclic high scores can partially cancel direction violations				

2026b); Structured Reasoning applies attention-MaxFlow or cross-response LCS rewards to explicitly tagged traces (Dong et al., 2026a); and the Graph Reasoning Paradigm makes the policy emit labeled graph nodes, reporting a 7,040-token AIME mean that exceeds our cap (Liu et al., 2026a). ARLCP and Self-Aligned Reward instead target efficient reasoning without explicit support direction (Yu et al., 2026; Han et al., 2026). Because their models, training, and decoders differ, these are external bars rather than rows mixed into the reproduced Pareto comparison.

4.3 WHERE THE GAIN COMES FROM (RQ2)

Raw direction and acyclicity. Table 2 uses policy ablations to test downstream utility and matched counterfactuals to test construct validity. Reversing the same steps ($n=120$) lowers q_{dir} by

Table 3: Matched aggregation controls with fixed sources and compute; Appendix Table 6 reports diagnostics and cost. **Cells await canonical outputs.**

Aggregation / graph control	Structural interaction	GSM8K	MATH	Olymp.	AIME	Macro	Tok.	Pareto $\uparrow$	$q_{\text{dir}}\uparrow$	$q_{\text{acyc}}\uparrow$	Hack $\%\downarrow$
Equal-weight sum	None	-	-	-	-	-	-	-	-	-	-
Tuned additive	None	-	-	-	-	-	-	-	-	-	-
Verifier-stratified (HERO-style) (Tao et al., 2026)	Correct/incorrect strata	-	-	-	-	-	-	-	-	-	-
Multiplicative	All-source product	-	-	-	-	-	-	-	-	-	-
Choquet w/o dir.-acyc. interaction	Other pairs retained	-	-	-	-	-	-	-	-	-	-
Topology-conditioned Choquet	Direction/acyclicity pairs	-	-	-	-	-	-	-	-	-	-
SARL-style graph reward	Undirected transitions	-	-	-	-	-	-	-	-	-	-

Table 4: Independent three-copy edge audit: recovery, topology validity, length robustness, and agreement. **Cells await annotation.**

Extractor	n	Overall edge recovery			Topology validity			Edge F1 by trace length			Agreement κ
		Prec.	Rec.	F1	Dir.	Cyc.	Cov.	Short	Medium	Long	
Adjacent-step chain	120	-	-	-	-	-	1.00	-	-	-	-
Rules only	120	-	-	-	-	-	-	-	-	-	-
Pair encoder (no abstention)	120	-	-	-	-	-	1.00	-	-	-	-
Frozen pair encoder (calibrated)	120	-	-	-	-	-	-	-	-	-	-
Semantic teacher (audit upper bound)	120	-	-	-	-	-	-	-	-	-	-
Projected DAG only	120	-	-	-	-	-	-	-	-	-	-
Hybrid raw-graph encoder	120	-	-	-	-	-	-	-	-	-	-

0.758 (paired bootstrap 95% interval $[0.702, 0.808]$), while q_{acyc} increases by 0.059 ($[0.034, 0.088]$) because reversal need not create circular support. In 56 pairs matched for edge count, backward mass, and incident coverage, only cycle closure changes: it creates a 0.339 acyclicity gap ($[0.299, 0.383]$) with zero direction-score difference. Noise controls are likewise selective: 29.5% direction flips reduce q_{dir} by 0.216, whereas 26.1% false-edge addition reduces q_{acyc} by 0.220 (Appendix Table 7 and Figure 3).

Separation matters: on reversed controls the scores are anticorrelated ($r = -0.483$, interval $[-0.812, -0.224]$), so averaging shrinks cross-trace standard deviation from 0.142 for direction to 0.063, a ratio of 0.443 ($[0.295, 0.501]$); 30.0% of traces move oppositely. Thus q_{dir} exposes textual-order inconsistency while q_{acyc} adds global self-support information, and premature averaging partially cancels their distinct responses. The policy panel tests whether this extra resolution improves the accuracy-length frontier.

Non-additive aggregation. Table 3 holds sources and compute fixed. The HERO-style control applies correctness-group normalization and variance-aware prompt weighting (Tao et al., 2026); the SARL-style control replaces directed support with an undirected small-world score. We support topology interactions only if they improve the Pareto frontier or reward hacking beyond additive, verifier-stratified, and multiplicative controls.

4.4 INDEPENDENT STRUCTURAL AUDIT AND COST (RQ3)

Three blinded copies balance order variants and produce pairwise agreement plus a strict-majority reference. The online audit exposes every graph and score calculation (Appendix Figure 4); Table 4 reports edge, direction, cycle, coverage, and incorrect-plausible-trace audits. Cost separates offline labeling and encoder adaptation from online reward latency, memory, graph pairs/s, and GRPO share; logs record the exact $4 \times L20$ allocation.

5 CONCLUSION

We presented TopoPRM, which converts backward and cyclic support into direction and acyclicity supervision before DAG projection, then reuses both signals for reward aggregation and distillation without graph-format decoding. **Limitations.** Structural signals do not prove correctness and depend on edge recovery; we therefore require independent edge, direction, cycle, agreement, and abstention audits. Semantic and executable verification remain future work.

ETHICS STATEMENT

The canonical study uses public mathematics benchmarks and collects no demographic attributes. Human annotators label directed support relations in reasoning traces; their identities are not part of the released annotations. An archival appendix table retains a prior private teacher-grading benchmark for continuity with the earlier submission, but it does not support the ICLR claims. **Before submission, that table requires documented authorization, privacy review, and release constraints; otherwise it will be removed from the submitted appendix.**

REPRODUCIBILITY STATEMENT

This draft is being audited for implementation-to-equation consistency and row-level result provenance. The submission version will point to the exact method definitions, configurations, raw-result manifests, and anonymized code bundle used to reproduce every retained result.

AI USE STATEMENT

Generative AI tools assisted with prior-review analysis, literature search and synthesis, hypothesis and experimental-control refinement, method implementation and code review, dataset cleaning, reproducibility auditing, scientific-figure editing, and manuscript drafting and revision. Before submission, the authors will manually inspect the cited literature, review code and generated artifacts, and verify every reported result against raw outputs. The authors take responsibility for the final claims, code, artifacts, and results; this disclosure will be updated if additional uses occur before submission.

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A IMPLEMENTATION DETAILS

A.1 HYPERPARAMETERS

Table 5 lists the key hyperparameters for each training stage.

Table 5: Hyperparameter summary for the three training stages.

Parameter	Value	Description
<i>Stage I: Supervised Fine-Tuning (SFT)</i>		
Base model	Qwen3.5-9B	–
Adapter	LoRA $r=64$, $\alpha=128$	all-linear modules
Learning rate	5×10^{-5}	cosine, 3% warmup
Epochs	3	–
Selected checkpoint	step 1,100	minimum held-out loss 0.381609; final step 1,734
Batch size	1/device, accum 8	global batch 32 on 4 GPUs
Max seq length	4096	–
Precision	bf16	–
<i>Stage II: GRPO with TopoPRM Rewards</i>		
Base model	SFT step 1,100	held-out-loss selection
Adapter	LoRA $r=64$, $\alpha=128$	all-linear modules
Learning rate	5×10^{-6}	cosine, 5% warmup
Gradient steps	300	–
Batch size	1/device, accum 4	global prompt batch 16 on 4 GPUs
Num generations	4 per prompt	for GRPO grouping
Temperature	0.9	rollout sampling
β (KL penalty)	0.04	–
Max completion length	4096	prompt capped at 512
Precision	bf16	–
<i>Stage III: Topology-Guided Distillation</i>		
Teacher model	Stage-II 9B	frozen during distillation
Student model	Qwen3.5-4B	canonical cross-scale run
Adapter	LoRA $r=64$, $\alpha=128$	all-linear modules
Learning rate	2×10^{-5}	cosine, 3% warmup
Epochs	1	–
Batch size	2/device, accum 4	global batch 32 on 4 GPUs
Student temperature	0.7	student-sampled trace
Teacher temperature	0.0	greedy revision target
Target budget	1024 tokens	fixed across methods
Target selection	answer, format, direction, acyclicity	threshold calibrated on validation
Max seq length	4096	–
Precision	bf16	–

A.2 REWARD CONFIGURATION

The Choquet reward in Eq. 3 uses the five independent sources (r_{out} , r_{fmt} , q_{dir} , q_{acyc} , q_{cont}) and the interaction set $\{(o, d), (o, a), (d, c), (a, c), (d, a)\}$. **The exact non-negative capacity coefficients are pending author confirmation and are not yet a training or paper claim.** Once locked, the coefficients will sum to one and remain fixed across the canonical run and its matched controls. The additive control uses the capacity’s Shapley source importance; the multiplicative control uses those same exponents. No length or reference-solution graph enters the canonical main reward because generated traces have no node alignment with a reference solution.

A.3 HARDWARE AND INFRASTRUCTURE

Canonical ICLR runs use $4 \times$ NVIDIA L20 GPUs. We report measured wall-clock time and peak memory from this configuration rather than extrapolating from the earlier A100 runs.

A.4 EVALUATION PROTOCOL

All public benchmark evaluations use the Transformers generation backend. Pass@1 is one sampled response per item from a single seed-0 evaluation pass; this is neither a multi-seed training result nor an average over repeated decodes. The shared decoder uses temperature 0.6, top- $p=0.95$, top- $k=20$, min- $p=0$, and repetition penalty 1.0. This matches Qwen3 and DeepSeek-R1’s reasoning-time recommendations and keeps the public and adapted rows comparable; it differs from Qwen3.5’s general-task recommendation only in using temperature 0.6 instead of 1.0. Generation explicitly enables the standard key-value cache even if a released checkpoint retains a training-time `use_cache=false` setting; this is recorded in the result manifest. The four primary mathematics benchmarks use the same 4,096-token completion cap and the same instruction to provide a step-by-step solution with the final answer in `\boxed{\}`. We retain each released chat template and documented control: GoT-R1, OpenReasoning-Nemotron, and AceMath receive a user-only task; MiMo uses an explicit empty system turn; the released Nemotron-Cascade template injects the terminal “/think” control; and the AceMath, Qwen-family, and Klear templates activate their released reasoning format without an evaluator-added prefix. No model receives an uncounted response prefix. For mathematics tasks, the evaluator verifies the last nonempty boxed answer, or a short terminal region after the last explicit final-answer marker, against the dataset target; it never searches an unsubmitted truncated derivation for a matching intermediate expression. The reference is passed first to the asymmetric `math_verify` verifier. Visible length is obtained by retokenizing the decoded response without special tokens. Pass@5 and maj@5, when run as secondary diagnostics, draw five samples from the same decoder and reuse this equivalence relation rather than raw string equality. The manifest stores the message-role protocol, prefix flag, decoder parameters, evaluation seed, batch size, model and adapter paths, dataset fingerprint, and full per-example responses; cached results are reused only when these protocol fields match.

B CHOQUET CAPACITY AND BASELINES

For non-negative Möbius coefficients summing to one, Eq. 3 is a normalized monotone 2-additive capacity (Choquet, 1954; Grabisch, 1997): it maps 0 to 0 and 1 to 1 and remains in $[0, 1]$. We separately replace the direction-acyclicity interaction with equal singleton mass to preserve both normalization and Shapley source importance, distinguishing the proposed structural synergy from merely increasing either source’s weight. We also test traces that are structurally plausible but answer-incorrect to detect reward hacking.

Matched baseline protocols. The LASER-D adaptation uses the true response-token count and four rollouts per prompt. It labels a prompt high, medium, or low by rollout accuracy thresholds $2/3$ and $1/3$, and adds 0.5 only to correct rollouts at or below the band’s target. Every 20 steps, the target is recomputed from a synchronized rolling monitor of 500 prompts. To match our 4,096-token cap, the released 4,096–16,384 candidate range is scaled by four to 1,024–4,096 in increments of 256; no character-length proxy is used. The HERO-style control holds the five source computations fixed: outcome is the binary verifier, while the Shapley-weighted and renormalized format, direction, acyclicity, and continuity sources form the dense score. We apply correct/incorrect-stratum min-max normalization with $\alpha=\beta=0.05$ and variance-aware prompt weights with $(w_{\min}, w_{\max}, k)=(0.4, 3.0, 6)$, matching HERO’s easy-to-verify mathematics setting (Tao et al., 2026). These are control rows, not the proposed aggregation.

C DAG EXTRACTION DETAILS

Extraction pipeline. Given a reasoning trace in `<think>`, we apply five frozen stages:

1. Step segmentation: split by numbered-step markers, mathematical discourse markers (`:`, `∴`), and line boundaries.
2. Sub-question partitioning: markers such as “Sub-question k ” define connected components; cross-sub-question edges are disallowed.
3. Information extraction: for each step, extract expressions, claims, and variables with rule-based patterns.

Table 6: Reward diagnostics for the matched aggregation runs in Table 3. Hack% is the rate of structurally plausible but answer-incorrect traces, zero-std groups are rollout groups with no reward contrast, and ms/trace is measured online reward latency. **Cells await canonical outputs.**

Aggregation / graph control	Hack%↓	Zero-std groups↓	ms/trace↓
Equal-weight sum	—	—	—
Tuned additive	—	—	—
Verifier-stratified (HERO-style) (Tao et al., 2026)	—	—	—
Multiplicative	—	—	—
Choquet w/o dir.-acyc. interaction	—	—	—
Topology-conditioned Choquet	—	—	—
SARL-style graph reward	—	—	—

4. Raw support encoding: combine high-precision symbolic candidates with a lightweight pair classifier that selects $i \rightarrow j$, $j \rightarrow i$, or abstention without imposing text order.

5. Projection: retain the raw graph for diagnostics, remove backward edges to obtain the training DAG, then add adaptive weak sequential edges only when support is absent.

Information-density filters. To avoid low-information nodes and noisy links: (a) claims are extracted at sentence level, with heading fragments and unfinished prefixes removed; (b) trivial tokens (single symbols, pure numerals, non-relational fragments) are excluded from dependency matching.

Dependency rules. For each unordered pair of steps in the same sub-question, rules propose candidates from informative expression reuse, canonicalized claim overlap, variable roles, and explicit step references. The pair classifier assigns direction or abstains. Only after raw-graph diagnostics are computed do we insert a fallback edge for an unsupported derivation or conclusion in the projected DAG.

Pair-encoder adaptation. The lightweight encoder is Qwen3.5-0.8B with a rank-8 LoRA three-way classification head. The frozen 9B semantic teacher provides 540 candidate source traces; deterministic malformed-segmentation filters retain 518, which are hash-split before order augmentation into 414 training and 104 validation sources. After original, reversed, and interleaved presentations, deterministic negative subsampling yields 10,854 training pairs; the unsampled validation set contains 3,419 pairs. On these teacher-held-out pairs, the selected epoch-3 encoder reaches 0.7873 three-class macro-F1, 0.7504 edge precision, 0.7893 edge recall, and 0.7603 direction accuracy on gold edges. **These are pre-correction parser diagnostics and are not final ICLR claims; the corrected-segmentation retraining is pending.** The calibrated edge-confidence and direction-margin thresholds are 0.40 and 0.10. This teacher-held-out diagnostic is not the independent human result in Table 4. The encoder is frozen during policy optimization; a retained encoder edge keeps its calibrated confidence as w_{ij} , while auditable rule edges use $w_{ij} = 1$. The adapter weights, corrected 120-trace audit pack, and all-order frozen prediction ledger have SHA-256 fingerprints ad8156c1...b6e, 95c440a3...b53, and a93be095...ec0, respectively; all order-control statistics are paired by source record.

Controlled edge noise. Table 7 measures how the two raw-graph diagnostics respond when frozen predictions are corrupted after extraction. Deletion probes missed edges, false-edge addition probes spurious support, and direction reversal probes orientation errors. The audit uses one deterministic perturbation draw per source trace and paired bootstrap across traces; it is not a repeated-training or multi-seed result.

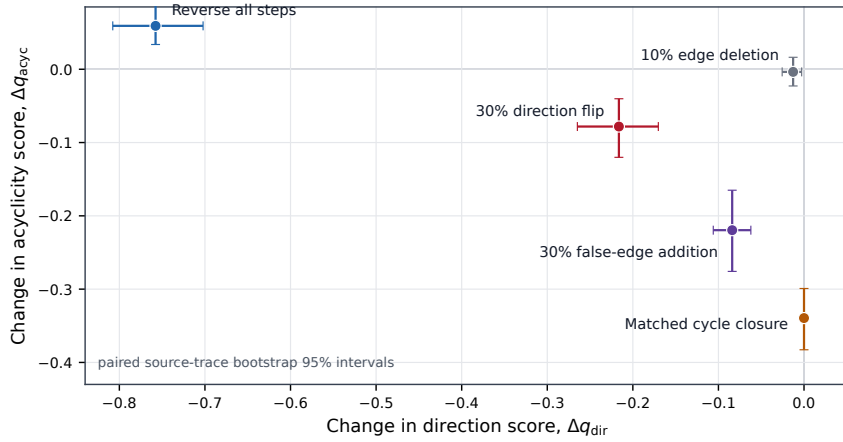


Figure 3: Controlled response of the frozen raw-graph encoder. Each point is the paired mean change relative to the identical source trace; bars are paired source-trace bootstrap 95% intervals. Reversal primarily changes direction, matched cycle closure changes acyclicity at fixed backward mass, and stochastic edge corruption moves both diagnostics according to its targeted failure. These are deterministic perturbation audits, not downstream policy results or training-seed variation.

Table 8: Diagnostic Best-of-8 selection accuracy (%) on one shared Qwen2.5-Math-7B-Instruct candidate pool ($n=80$ per benchmark; 2,048-token generation cap). PRM uses Qwen2.5-Math-PRM-7B; Hybrid multiplies normalized PRM by $1 + q_{\text{topo}}$. Oracle reports whether any candidate is correct. This diagnostic is separate from the primary 4,096-token policy comparison.

Benchmark	Pass@1	Maj@8	PRM@8	Topo@8	Hybrid@8	Oracle@8
GSM8K	93.75	95.00	95.00	91.25	95.00	96.25
MATH-500	83.75	85.00	83.75	78.75	83.75	92.50

Table 7: Frozen-encoder sensitivity on 120 source traces. One deterministic seed-0 perturbation is applied per trace; intervals are paired source-trace bootstrap 95% intervals. p_{real} is the realized modified-edge fraction. Deltas are corrupted minus original.

Perturbation	p	p_{real}	Δq_{dir}	Δq_{acyc}	$\Delta \text{Cov.}$	$\Delta \text{Cycle mass}$
Random edge deletion	.10	.072	-.013 [-.026, -.003]	-.004 [-.023, +.016]	-.030	-.011
	.20	.177	-.032 [-.054, -.011]	-.038 [-.061, -.015]	-.098	-.010
	.30	.284	-.066 [-.091, -.041]	-.061 [-.088, -.036]	-.153	-.018
Random false-edge addition	.10	.052	-.016 [-.025, -.008]	-.048 [-.071, -.028]	+.006	+.052
	.20	.136	-.043 [-.059, -.028]	-.097 [-.138, -.061]	+.018	+.109
	.30	.261	-.084 [-.106, -.062]	-.220 [-.276, -.165]	+.029	+.242
Random direction flip	.10	.100	-.067 [-.104, -.035]	-.027 [-.054, -.005]	.000	+.030
	.20	.207	-.142 [-.189, -.098]	-.047 [-.081, -.017]	.000	+.050
	.30	.295	-.216 [-.265, -.170]	-.078 [-.120, -.040]	.000	+.084

Topology is not a standalone correctness verifier. Table 8 rescales the rebuttal-era shared Best-of-8 pool with the canonical gold-first symbolic verifier. Topology-only selection underperforms the dedicated PRM on both subsets, and their simple product ties rather than improves on the PRM. We therefore use topology as an auditable training-time structural source and do not claim that it replaces semantic verification. The candidate-pool and cached-score ledgers have SHA-256 fingerprints 2f4ded08...2664 and 8304d8fc...b7c8.

TopoPRM trace audit

online topology verification · pre-correction · a93be0955b0

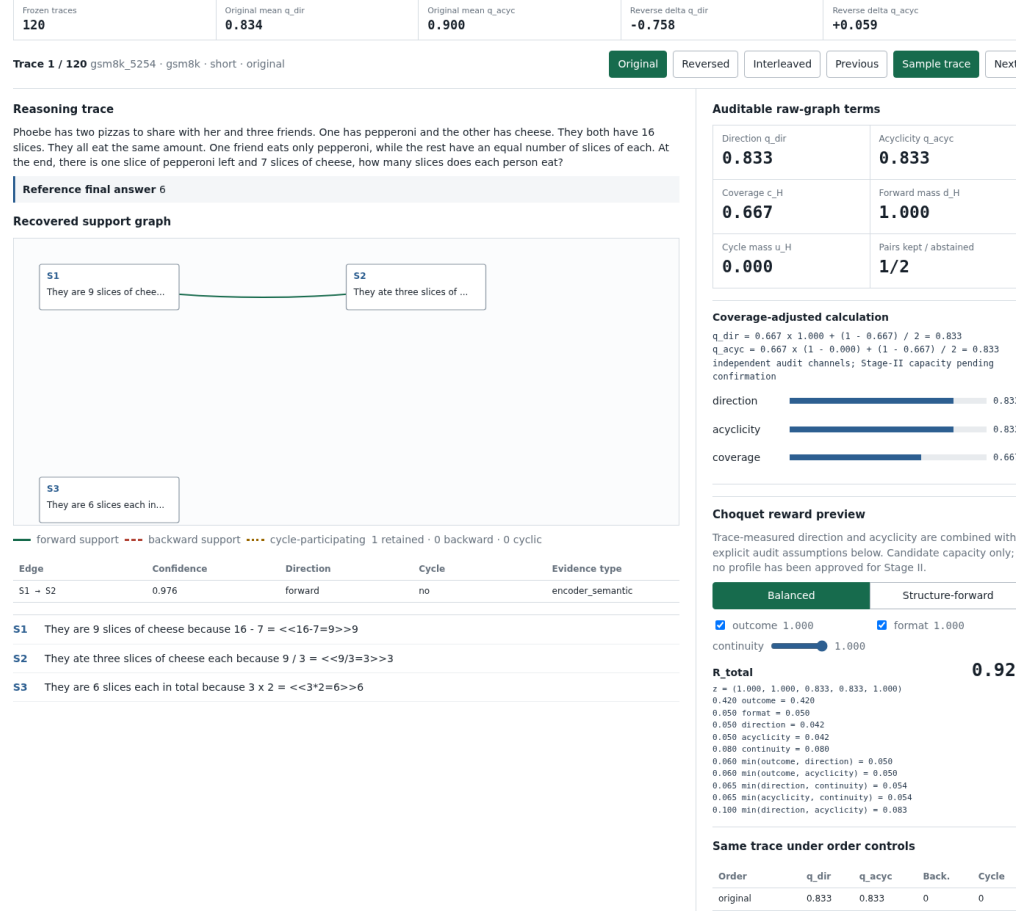


Figure 4: Trace-level topology audit interface. For each sampled reasoning trace, the dashboard exposes the recovered support graph, its edge ledger, and independently recomputed coverage-adjusted direction and acyclicity channels. The interactive reward preview substitutes these trace-measured channels into either preregistered candidate capacity while outcome, format, and continuity remain explicit user-controlled assumptions; it is not a canonical Stage-II result. Original, reversed, and interleaved controls switch the same trace online, and three linked annotation copies support the independent audit. The illustrated pre-correction encoder trace is one auditable sample, not an aggregate evaluation result.

Adaptive sequential weak edges. A weak sequential edge (weight 0.3) is added only when the target step lacks strong incoming support and the source step does not already emit strong outgoing dependencies. This prevents chain-dominant artifacts in long traces.

Node typing. Each node is classified into DEFINITION, DERIVATION, COMPUTATION, CONCLUSION, AUXILIARY, SUBSTITUTION, or CASE_ANALYSIS by keyword rules.

Relationship between DAG quality and reasoning compression. We test whether traces with stronger independent graph diagnostics (acyclicity, low orphan ratio, and directional consistency) admit shorter compressed forms under Topology-Guided Distillation. Intuitively, a well-structured DAG has fewer redundant branches and shorter critical paths, so the transitive reduction yields a more compact chain. The raw-graph scores q_{dir} and q_{acyc} enter Eq. 3 independently, while the projected DAG supplies localized revision targets. This relationship is evaluated rather than assumed.

D REWARD FUNCTION DETAILS

Outcome reward. Extracts the final mathematical answer from the last `\boxed{\}` expression, falling back to an explicit final numeric expression when needed. Symbolic equivalence to the reference answer is checked with `math_verify`; a correct answer receives 1 and an incorrect or unparseable answer receives 0.

Format reward. Scores recognizable mathematical reasoning and conclusion structure: a `<think>` block followed by a boxed answer receives 1.0; multi-step reasoning with a boxed answer receives 0.8; a boxed answer alone receives 0.5; a `<think>` block without a boxed conclusion receives 0.3; and unboxed step-by-step reasoning receives 0.1. Empty or unstructured output receives 0.

Length-only control. The explicit-efficiency baseline uses linear decay from 1.0 (at $\leq 2,000$ characters) to 0.0 (at $\geq 8,000$ characters). This signal is absent from the main TopoPRM reward; all evaluation tables independently report tokenizer-token counts.

Raw-graph rewards. The GRPO reward consumes the coverage-adjusted direction and acyclicity scores from Eqs. (1)–(2) as two aligned sources from one frozen encoder pass. It never replaces them with a scalar average. Orphan conclusions on the projected DAG remain an interpretable defect record for distillation but are not a sixth source in the canonical Stage-II capacity.

Continuity reward. Uses regex-based expression matching to verify that each step references expressions or claims from prior steps or the problem statement. The continuity score q_{cont} (Eq. 5) measures the fraction of supported steps.

E SUPPLEMENTARY MATHEMATICAL DEFINITIONS

Independent topology sources. Let c_H be the fraction of nodes incident to a dependency edge in the raw graph, and let d_H and u_H be the confidence-weighted forward-edge and cycle-edge mass defined in Eqs. (1)–(2). The affine shrinkage by c_H maps missing edge evidence to uncertainty at 0.5 rather than structural perfection. The two scores stay separate through aggregation: q_{dir} responds to local reversal without requiring a cycle, whereas q_{acyc} responds to strongly connected support even when backward mass is held fixed. Their interaction in Eq. 3 can reward joint satisfaction without cancelling their distinct counterfactual responses.

Continuity score (full form). The local continuity quality is

$$q_{\text{cont}}(y) = \begin{cases} 1.0, & \eta(y) = 1, \\ 0.8\eta(y), & \eta(y) < 1, \end{cases} \quad \eta(y) = \frac{1}{T} \sum_{i=1}^T \mathbb{I}[\text{supported}(s_i, s_{<i}, q)]. \quad (5)$$

Remark. Eq. (5) is the engineering-tuned continuity term used in the implementation. It is a separate source in the Choquet reward rather than being folded into topology before aggregation.

Topology-conditioned revision dispatch. The revision instruction is generated by

$$P_r = \psi(r_{\text{out}}, q_{\text{dir}}, q_{\text{acyc}}, \rho_{\text{orphan}}, q_{\text{cont}}, \ell), \quad (6)$$

where ℓ identifies the offending edge, cyclic component, or unsupported step when one can be localized.

F TOPOLOGY-GUIDED DISTILLATION DETAILS

Topology-Guided Distillation is an optional compression stage whose goal is to retain the teacher’s structural reasoning profile while reducing generation length in compact students. Table 9 compares static traces, generic revision, length-only revision, and Topology-Guided Distillation under the same teacher, student initialization, accepted-target count, and budgets. Direction and acyclicity retention must survive the independent audit; this extension returns to the main paper only if the single canonical run supports a central claim.

Table 9: Matched 9B-to-4B distillation protocol. All student methods use the same teacher, candidate prompts, accepted-target count, token budget, and optimizer budget. Accuracy is pass@1; Dir. and Acyc. are paired with the independent edge audit. **Cells remain pending until the single canonical run; marginal accuracy uses Wilson intervals and matched differences use paired item-level bootstrap intervals, neither training variance.**

Method	GSM8K	MATH-500	Avg. Tok	Dir.	Acyc.	Accept%
9B teacher	—	—	—	—	—	—
4B initialization	—	—	—	—	—	—
Static teacher traces	—	—	—	—	—	—
Generic revision	—	—	—	—	—	—
Length-only revision	—	—	—	—	—	—
Topology-Guided Distillation	—	—	—	—	—	—

Algorithm 1 Topology-Guided Distillation Training Pipeline

Require: Base policy π_0 ; data $\mathcal{D}_{\text{sft}}, \mathcal{D}_{\text{rl}}$; Choquet coefficients $\{m_i, m_{ij}\}$
Ensure: Teacher π_T , compact student π_S

- 1: $\pi_\theta \leftarrow \text{SFT}(\pi_0, \mathcal{D}_{\text{sft}})$ ▷ Stage I
- 2: **for** each GRPO step on $\mathcal{D}_{\text{rl}}$ **do** ▷ Stage II
- 3: Sample $\{y^{(i)}\}_{i=1}^G \sim \pi_\theta(\cdot | x)$; build $\mathcal{H}^{(i)} = \mathcal{E}_\phi(y^{(i)})$ and $\mathcal{G}^{(i)} = \Pi(\mathcal{H}^{(i)})$
- 4: Compute $r_{\text{out}}^{(i)}, r_{\text{fmt}}^{(i)}, q_{\text{dir}}^{(i)}, q_{\text{acyc}}^{(i)}, q_{\text{cont}}^{(i)}$
- 5: Compute $r_{\text{total}}^{(i)}$ by Eq. 3; normalize within the GRPO group
- 6: Update π'_θ with the GRPO objective
- 7: **end for**
- 8: $\pi_T \leftarrow \pi'_\theta$; init. π_S from a smaller pretrained checkpoint
- 9: **for** each prompt $x \in \mathcal{D}_{\text{rl}}$ **do** ▷ Stage III
- 10: $y_{\text{init}} \sim \pi_S(\cdot | x)$; extract defect record $F(y_{\text{init}})$ from $\mathcal{H}$ and $\mathcal{G}$
- 11: $P_r \leftarrow \psi(F(y_{\text{init}}))$; sample $y_{\text{rev}} \sim \pi_T(\cdot | x, y_{\text{init}}, P_r)$
- 12: Retain (x, y_{rev}) only if answer, format, direction, acyclicity, and budget constraints pass
- 13: Update π_S with $\mathcal{L}_{\text{distill}}$ (Eq. 4)
- 14: **end for**
- 15: **return** (π_T, π_S)

Why static trace distillation is insufficient. Conventional trace distillation (Hinton et al., 2015; Gu et al., 2024) presents every student with the same teacher solutions. It therefore cannot target a student’s particular missing dependency, and a length-only target can become shorter by deleting support that is necessary for the conclusion. Topology-Guided Distillation instead constructs supervision after observing the compact student’s own trace.

Defect localization and revision. For each initial student attempt, we retain the raw edge list and projected DAG. The defect record contains the endpoints of backward edges, nodes in a non-trivial strongly connected component, orphan conclusions in the projected DAG, and unsupported local transitions. The dispatcher selects the highest-priority localized defect and asks the frozen Stage-II teacher to rewrite the solution within a fixed response budget while preserving a boxed final answer.

Target acceptance. A revised trace is retained only if it has a correct boxed answer and valid format, reaches the calibrated topology threshold, does not reduce coverage-adjusted direction consistency or acyclicity relative to the initial trace, and stays within the shared token budget. We log every rejection reason and report acceptance coverage. This makes direction and acyclicity part of the distillation data contract, not merely an evaluation diagnostic.

Student training and controls. The 4B student is trained on accepted revisions with Eq. 4. Matched controls use the same teacher, prompts, number of accepted targets, and optimization budget but replace topology-conditioned dispatch with either a generic revision instruction or an explicit length-only instruction. External answer accuracy and generated tokens are primary; extractor-based

1296 structural retention is reported as internal-consistency evidence and paired with the independent hu-
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G RETAINED EMNLP SUBMISSION RESULTS

Archival evidence, not current ICLR claims. The nine result tables rendered in the EMNLP submission PDF are retained below with their numeric values unchanged. Retired method names are expanded or replaced only to avoid reintroducing abandoned terminology. These tables mix earlier checkpoints, decoders, response budgets, and in some cases explicitly marked approximate entries. The ICLR main tables use only the canonical single-run protocol in Section 4.1; archival rows will be promoted only after row-level provenance and protocol revalidation.

Table 10: Archived EMNLP Table 1: results on nine public benchmarks across four task categories (Elementary: GSM8K, MATH-500; Olympiad: OlympiadBench, Omni-Math; Competition: AIME’24, AIME’25, CNMO’24; General: MMLU, GPQA-Diamond). All numbers are the originally submitted pass@5 accuracies (%).

Model	Params	Elementary		Olympiad		Competition			General		Avg.
		GSM8K	MATH	Olympiad	Omni	AIME'24	AIME'25	CNMO'24	MMLU	GPQA-D	
Reference open-source models											
Qwen3.5-2B	2B	77.2	60.8	48.5	13.3	6.7	13.3	36.1	32.8	37.9	36.3
Qwen3.5-4B	4B	85.1	55.8	32.8	43.4	0.0	13.3	10.0	67.4	29.8	37.5
Qwen2.5-Math-7B-Instruct (Yang et al., 2024b)	7B	95.2	73.8	38.0	22.0	13.0	10.0	13.3	54.8	28.3	38.7
Llama-3.1-8B-Instruct (Grattafiori et al., 2024)	8B	84.5	52.2	15.0	10.0	3.0	2.0	3.0	68.0	30.0	29.7
DeepSeek-R1-Distill-Llama-8B (DeepSeek-AI, 2025)	8B	83.0	89.1	48.0	25.0	50.4	35.0	25.0	52.0	49.0	50.7
Qwen2.5-7B (Base) (Yang et al., 2024a)	7B	85.4	49.8	25.0	18.0	7.0	5.0	8.0	74.2	36.4	34.3
+ SFT	7B	91.3	65.4	42.5	54.2	16.7	12.2	16.7	70.1	28.3	44.2
+ SimpleRL-Zoo (Zeng et al., 2023)	7B	91.7	78.2	38.6	40.4	20.0	15.6	15.0	62.5	55.2	46.4
+ Open-Reasoner-Zero (Hu et al., 2026)	7B	89.0	81.4	47.9	38.4	17.9	15.6	13.3	62.5	36.6	44.6
+ TopoPRM (Ours)	7B	95.8	72.6	47.8	57.2	16.7	26.7	47.0	82.7	51.0	55.3
DeepSeek-R1-Distill-Qwen-7B (Base) (DeepSeek-AI, 2025)	7B	73.8	36.8	25.8	28.0	46.7	30.0	22.9	23.7	12.6	33.4
+ SFT	7B	86.0	92.8	52.0	48.0	55.5	40.0	30.0	50.0	49.1	55.9
+ GRPO (Shao et al., 2024)	7B	85.1	67.4	56.2	72.0	50.0	40.0	57.8	50.1	17.7	55.1
+ TopoPRM (Ours)	7B	84.3	66.6	56.4	72.2	55.5	40.0	55.4	60.3	34.3	58.3
Qwen3.5-9B (Base)	9B	69.5	66.6	55.2	61.5	16.7	21.1	16.7	28.6	37.4	41.5
+ SFT	9B	94.1	68.2	44.0	53.8	30.0	15.6	30.0	78.4	47.0	51.2
+ GRPO (Shao et al., 2024)	9B	93.3	63.0	56.2	72.0	46.7	30.0	57.8	63.0	17.7	55.5
+ TopoPRM (Ours)	9B	97.9	81.9	56.2	76.2	43.3	18.9	47.0	86.4	40.4	60.9

Table 13: Archived EMNLP Table 9: reward aggregation comparison on the in-domain critique set. All numeric values are retained from the submission.

Strategy	Mid	High	Fmt%	Len	Collapse%↓
Linear (weighted sum)	37.2	22.3	94.6	364	75.1
Clipped linear	29.0	21.5	93.0	382	67.7
Legacy hierarchical aggregation	37.2	22.3	94.6	364	37.9

Table 11: Archived EMNLP Table 3: reward component ablation on Qwen3.5-9B. Avg is the originally submitted unweighted mean pass@1 across eight benchmarks.

Reward Config.	Avg (%)	Δ	Tok	Acc/kTok $\uparrow$	Collapse% $\downarrow$
TopoPRM (Full)	45.6	—	1692	26.9	37.9
w/o ACE	41.7	−3.8	1958	21.3	42.1
w/o Topology	41.8	−3.8	1943	21.5	44.6
w/o Continuity	16.3	−29.3	2047	8.0	68.8
Outcome Only	40.8	−4.7	1823	22.4	57.8

Table 12: Archived EMNLP Table 4: cross-scale distillation from the 9B TopoPRM teacher. Token Ratio is mean generation length relative to the teacher; Structural Retention aggregates the originally submitted DAG-profile metrics on MATH-500.

Scale	GSM8K	MATH	Tok	Ratio	Struct. Ret.
9B (Teacher)	84.3	66.6	679	1.00×	1.00×
4B	82.8	61.5	382	0.56×	0.93×
2B	78.9	57.1	305	0.45×	0.88×
0.8B	73.6	52.4	244	0.36×	0.81×
4B (SFT-distill)	79.4	58.2	438	0.64×	0.86×
4B (Off-policy KL)	76.8	49.3	4096	>1.0×	0.41×

Table 14: Per-benchmark results and component ablations of TOPOPRM. We report **p@1**, **p@5**, **maj@5**, mean generated tokens / total wall-clock seconds (**Tok/Time**, in ks), and token-efficiency **Acc/kTok** = p@1/(tokens/1000). We use the same protocol in A. **Bold** / underline: best / second-best per column.

Dataset/Method	GSM8K (1,319)					MATH-500 (500)					OlympiadBench (134)					Omni-MATH (262)				
	p@1	p@5	maj@5	Tok/Time	Acc/kTok	p@1	p@5	maj@5	Tok/Time	Acc/kTok	p@1	p@5	maj@5	Tok/Time	Acc/kTok	p@1	p@5	maj@5	Tok/Time	Acc/kTok
Base (Qwen3.5-9B)	55.2	87.8	55.7	1536/45.0	35.9	19.8	39.2	21.4	1536/18.7	12.9	11.2	25.4	14.2	2560/26.5	4.4	16.4	33.2	17.9	2559/52.0	6.4
Outcome Only	93.3	97.7	93.6	1047/34.2	89.1	<u>50.8</u>	59.0	55.2	1511/18.9	33.6	31.3	46.3	37.3	<u>2043/19.7</u>	<u>15.3</u>	41.6	<u>55.3</u>	48.9	<u>1845/35.2</u>	<u>22.5</u>
w/o Topology	93.0	97.7	93.9	<u>946/25.0</u>	<u>98.3</u>	<u>51.0</u>	<u>59.4</u>	55.6	<u>1506/16.2</u>	<u>33.9</u>	29.8	<u>50.7</u>	<u>38.1</u>	2468/11.8	12.1	<u>42.4</u>	54.2	45.8	2430/17.8	17.4
w/o Continuity	51.0	86.1	52.6	1528/48.3	33.4	21.6	39.6	23.2	1536/19.1	14.1	8.2	25.4	9.0	2560/ 8.7	3.2	14.9	30.5	17.6	2560/18.5	5.8
w/o ACE	<u>93.8</u>	98.0	<u>94.6</u>	1006/28.3	93.2	<u>50.8</u>	<u>59.8</u>	<u>55.2</u>	1515/18.2	33.5	32.8	47.0	35.8	2491/ 9.6	13.2	42.0	53.8	47.3	2463/19.4	17.1
★ TopoPRM (Full)	94.1	<u>97.9</u>	94.8	796/19.7	118.2	51.0	81.9	54.6	1454/16.6	35.1	32.8	50.7	38.8	1768/18.1	18.6	42.8	55.7	<u>47.7</u>	1592/19.0	26.9

Dataset/Method	AIME 2024 (30)					AIME 2025 (90)					CNMO 2024 (30)					MMLU (500)				
	p@1	p@5	maj@5	Tok/Time	Acc/kTok	p@1	p@5	maj@5	Tok/Time	Acc/kTok	p@1	p@5	maj@5	Tok/Time	Acc/kTok	p@1	p@5	maj@5	Tok/Time	Acc/kTok
Base (Qwen3.5-9B)	16.7	33.3	33.3	4083/ 8.5	4.1	2.2	4.4	1.1	2560/ 5.9	0.9	10.0	16.7	3.3	2560/ 1.9	3.9	28.6	41.4	26.2	512/55.1	55.9
Outcome Only	16.7	33.3	20.0	<u>2546/ 5.6</u>	6.6	<u>13.3</u>	15.6	8.9	<u>2547/16.7</u>	<u>5.2</u>	16.7	<u>33.3</u>	<u>26.7</u>	<u>2530/ 5.5</u>	6.6	63.0	84.0	61.0	512/146.2	123.0
w/o Topology	<u>20.0</u>	<u>40.0</u>	<u>30.0</u>	2560/ 2.1	<u>7.8</u>	12.2	<u>16.7</u>	10.0	2560/ 6.2	4.8	<u>20.0</u>	40.0	<u>26.7</u>	2560/ 2.0	<u>7.8</u>	<u>65.8</u>	<u>85.0</u>	<u>63.0</u>	512/32.3	<u>128.5</u>
w/o Continuity	6.7	13.3	3.3	2560/ 2.3	2.6	1.1	4.4	0.0	2560/ 7.0	0.4	6.7	20.0	3.3	2560/ 2.4	2.6	20.2	34.8	18.6	512/80.4	39.5
w/o ACE	<u>20.0</u>	30.0	23.3	2560/ 2.6	<u>7.8</u>	<u>13.3</u>	<u>17.8</u>	12.2	2560/ 7.7	<u>5.2</u>	<u>20.0</u>	30.0	<u>26.7</u>	2560/ 2.6	<u>7.8</u>	61.1	83.1	61.2	508/58.1	120.3
★ TopoPRM (Full)	30.0	43.3	33.3	2431/ 5.4	12.3	15.6	18.9	<u>11.1</u>	2457/ 7.7	6.4	30.0	47.0	30.0	2530/ 2.5	11.9	68.2	86.4	68.8	508/48.0	134.2

Table 15: Main results on the in-domain mathematical critique benchmark. **Middle** and **High** denote middle-school and high-school subsets. Acc measures critique scoring accuracy; Fmt% measures structural compliance; Len reports mean prediction length in tokens. Best results per column are in **bold**, second-best are underlined.

Model	Middle School			High School			Overall
	Acc	Fmt%	Len	Acc	Fmt%	Len	
Zero-shot Baseline							
Qwen3-32B	16.0	64.5	1 200	7.0	43.5	1 240	11.5
Qwen3-32B + SFT	29.0	78.5	1 164	19.5	65.0	1 233	24.3
RFT Variants							
+ Outcome Only	19.5	89.0	410	13.0	85.5	475	16.3
+ w/o Topology	23.0	91.0	397	16.5	86.0	460	19.8
+ w/o Continuity	27.5	90.5	379	20.0	85.0	436	23.8
+ Clipped Linear	29.0	93.0	382	21.5	88.5	458	25.3
+ TopoPRM (full)	37.2	94.6	364	22.3	89.3	424	29.7
Topology-Guided Distillation Fine-tuned							
Qwen3.5-9B + full	40.5	98.0	1 602	30.5	94.0	2 191	35.5
Qwen3-8B + full	60.0	94.0	532	58.0	97.0	613	59.0

Table 16: Reward component ablation on **Qwen3-32B** over the in-domain critique validation set. Each configuration toggles the topology (r_{topo}) and continuity (r_{cont}) signals on top of the outcome reward r_{out} . **Avg** is the unweighted mean of Middle- and High-school accuracy; Δ denotes the gap relative to the full TopoPRM. Rows are sorted by Avg in ascending order to highlight the monotonic contribution of each component.

Configuration	Reward Components			Accuracy (%)			Δ
	r_{out}	r_{topo}	r_{cont}	Mid	High	Avg	
Outcome Only	✓	✗	✗	19.5	13.0	16.3	−13.0
w/o Topology	✓	✗	✓	23.0	16.5	19.8	−9.5
w/o Continuity	✓	✓	✗	27.5	20.0	23.8	−5.5
TopoPRM (Full)	✓	✓	✓	33.0	25.5	29.3	—

Table 17: Accuracy-length trade-off on the in-domain critique validation set (Qwen3-32B). Acc/k-Tok = Overall-Acc per 1,000 tokens. Len $\uparrow$ is the length increase relative to TopoPRM (lower is better).

Config.	Middle		High		Efficiency	
	Acc	Len	Acc	Len	Acc/kTok $\uparrow$	Len $\uparrow$
TopoPRM (full)	37.2	364	22.3	424	75.5	—
w/o Continuity	27.5	379	20.0	436	58.4	+4.1%
w/o Topology	23.0	397	16.5	460	46.2	+9.0%
Outcome Only	19.5	410	13.0	475	36.8	+12.6%
Qwen3-32B (base)	16.0	1200	7.0	1240	9.4	+230%

Table 18: Structural quality of extracted reasoning DAGs on the in-domain middle-school evaluation set. Acyclic% is the fraction of valid DAGs; No-Orphan% is the fraction of traces without unsupported conclusion nodes; Dir-Cons. is mean directional consistency; DAG Depth is mean dependency depth; Edge-Keep% is the fraction of dependency edges retained after compression.

Model	Acyclic% $\uparrow$	No-Orphan% $\uparrow$	Dir-Cons. $\uparrow$	DAG Depth	Edge-Keep% $\uparrow$
Qwen3-32B + SFT	100	61	0.49	1.05	43
+ GRPO (outcome only)	100	82	0.89	1.12	56
+ GRPO (w/o topology)	100	84	0.91	1.14	58
+ GRPO (w/o continuity)	100	83	0.91	1.13	57
+ GRPO (TopoPRM full)	100	89	1.00	1.17	60

G.1 ADDITIONAL EMNLP-WORKSPACE RESULTS

The following six supplementary tables were present in the accompanying EMNLP source workspace but were not among the nine result tables rendered in the submission PDF. Their values are retained for reference under the same archival restrictions.

Table 19: Accuracy evaluation across 9 public math reasoning benchmarks. All our models follow one transformers-backend protocol with the model’s native chat template, shared answer extraction, and shared matching. Open-reported numbers are taken from official papers and model cards. **Bold** marks the best result per block and underline marks the second best. **This is the retained EMNLP-submission table: entries prefixed by ~ are archival approximations, not ICLR claims, and all rows are being revalidated under the canonical protocol.**

Model	Params	Elem. Math		Olympiad Math		Competition		CNMO	General	
		GSM8K	MATH-500	Olympiad	Omni-MATH	AIME'24	AIME'25	CNMO'24	MMLU	GPQA-D
Reference open-source models (open-reported)										
Qwen2.5-7B-Instruct (Yang et al., 2024a)	7B	91.6	75.5	~25.0	~18.0	~7.0	~5.0	~8.0	75.4	36.4
Qwen2.5-Math-7B-Instruct (Yang et al., 2024b)	7B	95.2	83.6	~38.0	~22.0	~13.0	~10.0	~15.0	~60.0	~35.0
Llama-3.1-8B-Instruct (Grattafiori et al., 2024)	8B	84.5	52.2	~15.0	~10.0	~3.0	~2.0	~3.0	~68.0	~30.0
DeepSeek-R1-Distill-Llama-8B (DeepSeek-AI, 2025)	8B	~83.0	89.1	~48.0	~25.0	50.4	~35.0	~25.0	~52.0	49.0
DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025)	7B	~86.0	92.8	~52.0	~28.0	55.5	~40.0	~30.0	~55.0	49.1
SimpleRL-Zoo (Qwen2.5-7B) (Zeng et al., 2023)	7B	91.7	78.2	~38.6	~40.4	20.0	~15.6	~15.0	~62.5	~55.2
Ours: 9B family (Qwen3.5-9B base, unified transformers protocol)										
Qwen3.5-9B (base)	9B	67.1	43.8	29.8	51.4	6.7	3.3	12.1	52.9	27.3
+ SFT	9B	94.1	50.8	32.8	42.4	30.0	15.6	30.0	68.2	20.7
+ GRPO (outcome-only)	9B	93.3	50.8	31.3	41.6	16.7	13.3	16.7	63.0	–
+ GRPO (w/o topology)	9B	93.0	51.0	29.8	42.4	20.0	12.2	20.0	65.8	16.2
+ GRPO (w/o continuity)	9B	–	–	–	–	–	–	–	–	–
+ TopoPRM (hierarchical)	9B	93.5	49.8	32.8	42.8	26.7	12.2	26.7	61.8	16.7
+ TopoPRM (gated)	9B	93.8	50.8	32.8	42.0	20.0	13.3	20.0	61.1	21.2
Ours: DR1-7B family (DeepSeek-R1-Distill-Qwen-7B base, unified transformers protocol)										
DR1-7B (base, measured)	7B	60.8	68.4	57.8	72.8	46.7	33.3	23.3	42.4	16.2
+ SFT	7B	73.8	36.8	25.8	48.0	0.0	0.0	22.9	23.7	12.6
+ GRPO (outcome-only)	7B	85.1	67.4	56.2	72.0	46.7	30.0	57.8	50.1	17.7
+ GRPO (w/o topology)	7B	84.5	68.8	–	–	36.7	33.3	53.0	–	11.1
+ GRPO (w/o continuity)	7B	85.1	66.4	–	–	36.7	26.7	50.6	–	14.6
+ TopoPRM (hierarchical)	7B	84.3	66.6	56.4	72.2	50.0	40.0	55.4	49.6	15.2
Ours: Distilled students (Topology-Guided Distillation, target ≤4B)										
Student (4B, SFT distill)	4B	~89.4	~49.8	~24.0	~30.2	~12.1	~8.5	~6.8	~82.6	~40.7
Student (4B, Topology-Guided Distillation)	4B	~91.3	~52.1	~27.6	~34.8	~15.4	~10.2	~8.1	~83.5	~41.9
Student (2B, Topology-Guided Distillation)	2B	~86.7	~46.5	~21.3	~27.1	~9.2	~6.4	~5.7	~80.8	~38.4
Student (0.8B, Topology-Guided Distillation)	0.8B	~79.8	~39.4	~14.0	~18.6	~6.1	~4.5	~3.9	~76.3	~34.2

Table 20: Detailed unified metrics across our measured configurations (num_samples=5, native chat template, same extractor). **Cor** = correct count; **Err** = error count; **Tok** = mean generation tokens; **Acc/kTok** = pass@1 points per 1 000 tokens (higher is more token-efficient). Best within each block in **bold**.

GSM8K									
Model	Params	pass@1	pass@5	maj@5	prm@5	F1	Cor/Err	Tok	Acc/kTok
<i>DRI-7B family</i>									
DR1-7B (base, measured)	7B	60.8	80.8	64.2	60.8	60.8	800/519	617	98.5
+ SFT	7B	73.8	91.5	80.2	73.7	73.8	970/349	634	116.4
+ GRPO (outcome-only)	7B	85.1	94.0	88.3	85.1	85.1	1120/199	674	126.3
+ GRPO (w/o topology)	7B	84.5	93.9	87.5	84.5	84.5	1113/206	686	123.2
+ GRPO (w/o continuity)	7B	85.1	94.2	87.9	85.1	85.1	1121/198	640	133.0
+ TopoPRM (hierarchical)	7B	84.3	93.6	88.3	84.3	84.3	1110/209	679	124.2
<i>9B family (our unified protocol)</i>									
Qwen3.5-9B (base)	9B	67.1	—	—	—	67.1	850/469	1536	43.7
MATH-500									
Model	Params	pass@1	pass@5	maj@5	prm@5	F1	Cor/Err	Tok	Acc/kTok
<i>DRI-7B family</i>									
DR1-7B (base, measured)	7B	68.4	70.6	67.4	68.4	68.4	322/178	4957	13.8
+ SFT	7B	36.8	54.8	43.4	36.8	36.8	185/315	3731	9.9
+ GRPO (outcome-only)	7B	67.4	72.6	67.4	67.4	67.4	318/182	6001	11.2
+ GRPO (w/o topology)	7B	68.8	73.2	68.8	68.8	68.8	324/176	5973	11.5
+ GRPO (w/o continuity)	7B	66.4	71.8	67.8	66.4	66.4	312/188	6088	10.9
+ TopoPRM (hierarchical)	7B	66.6	72.8	68.2	66.6	66.6	314/186	6053	11.0
<i>9B family (our unified protocol)</i>									
Qwen3.5-9B (base)	9B	43.8	—	—	—	43.8	203/297	3072	14.3
AIME 2024									
Model	Params	pass@1	pass@5	maj@5	prm@5	F1	Cor/Err	Tok	Acc/kTok
<i>DRI-7B family</i>									
DR1-7B (base, measured)	7B	46.7	46.7	36.7	46.7	46.7	10/20	8138	5.7
+ SFT	7B	0.0	3.3	0.0	0.0	0.0	0/30	4301	0.0
+ GRPO (outcome-only)	7B	46.7	36.7	33.3	46.7	46.7	9/21	8192	5.7
+ GRPO (w/o topology)	7B	36.7	46.7	33.3	36.7	36.7	8/22	8192	4.5
+ GRPO (w/o continuity)	7B	36.7	40.0	33.3	36.7	36.7	8/22	8192	4.5
+ TopoPRM (hierarchical)	7B	50.0	40.0	33.3	50.0	50.0	10/20	8192	6.1
<i>9B family</i>									
Qwen3.5-9B (base)	9B	6.7	—	—	—	6.7	1/29	4096	1.6
Omni-MATH (500 subset)									
Model	Params	pass@1	pass@5	maj@5	prm@5	F1	Cor/Err	Tok	Acc/kTok
<i>DRI-7B family</i>									
DR1-7B (base, measured)	7B	72.8	—	—	72.8	72.8	348/152	4720	15.4
+ SFT	7B	48.0	—	—	48.0	48.0	235/265	7114	6.7
+ GRPO (outcome-only)	7B	72.0	—	—	72.0	72.0	347/153	3562	<u>20.2</u>
+ TopoPRM (hierarchical)	7B	72.2	—	—	72.2	72.2	348/152	3529	20.5

Table 21: Ablation of TopoPRM reward components on the Qwen3.5-9B backbone. All variants start from the same SFT checkpoint and run identical GRPO steps. We report pass@1 / pass@5 (%), mean generation tokens (Tok), and Acc/kTok = pass@1 per 1 000 tokens. Δ shows the change in pass@5 relative to the full TopoPRM (hierarchical). Best in **bold**.

Reward Config	GSM8K				MATH-500				AIME'24				Omni-MATH				MMLU			
	p@1	p@5	Tok	Δ p@5	p@1	p@5	Tok	Δ p@5	p@1	p@5	Tok	Δ p@5	p@1	p@5	Tok	Δ p@5	p@1	p@5	Tok	Δ p@5
TopoPRM (full, hier.)	93.5	97.9	1026	—	49.8	59.0	1506	—	26.7	43.3	2560	—	42.8	55.7	2465	—	61.8	81.9	508	—
TopoPRM (gated)	93.8	<u>97.9</u>	1006	+0.1	50.8	59.8	1515	+0.8	20.0	30.0	2560	−13.3	42.0	53.8	2463	−1.9	61.1	83.1	508	+1.3
w/o Topology	93.0	97.7	946	−0.2	51.0	59.4	1506	+0.4	20.0	40.0	2560	−3.3	42.4	54.2	2430	−1.5	65.8	<u>85.0</u>	512	+3.1
w/o Continuity	51.0	86.1	1528	−11.8	21.6	39.6	1536	−19.4	6.7	13.3	2560	−30.0	14.9	30.5	2560	−25.2	20.2	34.8	512	−47.1
Outcome Only GRPO	93.3	97.7	1047	−0.2	50.8	59.0	1511	+0.0	16.7	33.3	2546	−10.0	41.6	55.3	1845	−0.4	63.0	84.0	512	+2.1
SFT baseline	94.1	97.7	796	−0.2	50.8	<u>60.4</u>	1454	+1.4	<u>30.0</u>	43.3	2431	+0.0	<u>42.4</u>	53.1	1592	−2.7	<u>68.2</u>	86.4	512	+4.5

Table 22: Generation efficiency of Qwen3.5-9B variants. All runs use 5 samples per item on the same hardware. **Tok** = mean generation tokens per sample; **Time** = total wall-clock seconds for the benchmark; **Tok/s** = throughput (total tokens / time); **Acc/kTok** = pass@1 per 1 000 tokens (higher = more token-efficient). Best in **bold**.

Model	GSM8K (1,319 items)				MATH-500				AIME'24 (30 items)				Omni-MATH (262 items)			
	Tok	Time	Tok/s	Acc/kTok	Tok	Time	Tok/s	Acc/kTok	Tok	Time	Tok/s	Acc/kTok	Tok	Time	Tok/s	Acc/kTok
TopoPRM (Full)	796	19.7k	266	118.2	1454	16.6k	220	34.9	2431	5.4k	67	<u>12.3</u>	1592	31.7k	66	26.6
Outcome Only GRPO	1047	34.2k	202	89.1	1511	18.9k	200	33.6	2546	5.6k	68	6.5	1845	35.2k	69	22.5
w/o Topology	946	25.0k	250	98.3	1506	16.2k	232	33.9	2560	2.1k	186	7.8	2430	17.8k	179	17.4
w/o Continuity	1528	48.3k	209	33.4	1536	19.1k	201	14.1	2560	2.3k	165	2.6	2560	18.5k	181	5.8
Legacy gated agg.	1006	28.3k	234	93.2	1515	18.2k	208	33.5	2560	2.6k	147	7.8	2463	19.4k	167	17.0
TopoPRM (Hier.)	1026	33.4k	203	91.1	1506	18.0k	209	33.1	2560	2.6k	149	10.4	2465	19.0k	170	<u>17.3</u>

Table 23: First-Attempt vs Revised accuracy on 1K sampled AIME-2024 / MATH problems (Qwen3.5-9B family). First Attempt: one sample from the model. Revised Attempt: a second sample conditioned on the first attempt and a topology-aware revision prompt P_r (Eq. 6). Baseline = plain distillation reviser without DAG signal. Avg-Len refers to the revised response only.

Model	First Att.	Revised Att.	Gain	Revised Len	Revised / First
Qwen3.5-9B (base, no SRT)	~31.8	~34.6	~+2.8	~742	~1.07
+ Plain distillation reviser (no topology)	~33.1	~37.0	~+3.9	~688	~1.02
+ Topology-Guided Distillation reviser (Eq. 6)	~33.0	~39.8	~+6.8	~624	~0.94

Table 24: Case study comparing teacher and topology-guided distilled-student traces on the same prompt.

	Teacher (TopoPRM)	Distilled student
Problem	High-school algebra problem requiring multi-step substitution and verification (same prompt hash for both rows).	
# Steps	~14	~9
# Tokens	~612	~428
Score-Acc	~0.93	~0.91
Key reasoning	Core conclusion supported by an explicit minimal chain of derivations and substitutions.	Preserves key support edges while compressing redundant branches and intermediate restatements.